Stand delineation & HBAD models

Formulation and inputs — equations as implemented

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Stand delineation (GA + FL) — inputs and land status

Inputs (all on the TreeMap 30 m EPSG:5070 grid; 1 px = 0.09 ha)

- TreeMap 2022 plot ids + attributes (BA, height, QMD, cover, TPA, softwood share)
- Hansen GFC v1.12: loss year, tree cover 2000, land mask
- LANDFIRE EVT 2022: excluded types; tree / shrub / herb life form
- AlphaEarth 2022, 2025: 64-d embeddings, de-quantised $\mathrm{sign}(q)(q/127.5)^2$
- SOLUS 0–30 cm: clay, sand, SOC, AWC, pH, restrictive depth
- Barriers (1-px lines): TIGER roads and water, FL parcels

Status classes (recent = Hansen loss ≥ 2017; former forest = cover_{2000} ≥ 25%)

- **1 forest**: any TreeMap pixel
- **2 harvested**: recent loss on former forest, EVT not excluded (overrides 1)
- **3 regenerating**: not TreeMap, not excluded, and older loss, or a disturbed / plantation EVT, or shrub/herb with cover_{2000} ≥ 50%
- **4 forest unmapped**: EVT tree life form, not TreeMap
- **9 converted**: loss on former forest, now an excluded EVT (not forest land)
- 0 outside the AOI and on Hansen non-land

Speckle cleaning: same-class patches < 2 ha take the class of the nearest larger forest-land patch (≤ 15 px).

scripts 10_forest_land.py, 11_land_clean.py, config/pilot.yml

Stand delineation — the per-pixel feature vector (34 dimensions)

Each group is projected and scaled so that its total variance on a fit sample (60 tiles × 4000 forest-land pixels) equals w^2:

$$\mathbf{y}_{\text{AEF}} = (\mathbf{a}_{25} - \mu)\, V_{16}^{\square}\ 1.0\ /\sqrt{\textstyle\sum_{k\le 16} \lambda_k} \qquad \text{16 PCs, AlphaEarth 2025}$$

$$\mathbf{y}_{\text{old}} = (\mathbf{a}_{22} - \mu)\, V_{8}^{\square}\ 0.4\ /\sqrt{\textstyle\sum_{k\le 8} \lambda_k} \qquad \text{AlphaEarth 2022 on the 2025 basis (8)}$$

$$y_{\text{chg}} = \left(1 - \cos(\mathbf{a}_{25}, \mathbf{a}_{22})\right) 0.5\ /\ \mathrm{sd}(1 - \cos) \qquad \text{change since 2022 (1)}$$

$$\mathbf{y}_{\text{TM}} = (\mathbf{x} - \mu)/\sigma\ \cdot\ 0.6/\sqrt{6} \qquad \text{TreeMap attributes (6)}$$

$$\mathbf{y}_{\text{soil}} = \mathbf{z}\, V_{3}^{\square}\ 0.3\ /\sqrt{\textstyle\sum_{k\le 3} \lambda_k} \qquad \text{3 PCs of the z-scored soils}$$

- TreeMap: ln BA, height, ln QMD, cover, ln TPA, softwood share; zeroed (= sample mean) off TreeMap forest or where harvested.
- The weights are per-group standard deviations: AlphaEarth dominates, TreeMap and soils refine.

scripts/features.py (Featurizer.features), 20_fit_features.py

Stand delineation — superpixels and the region-adjacency merge

1. Superpixels — SLIC per 2048-px tile and per status, on forest land minus barrier pixels: $n_{seg} = \lfloor n_{px}/11 \rfloor$ (1 ha), compactness 0.3, connectivity enforced; never cross barriers, statuses or tiles.

2. Merge on the region-adjacency graph (edges = shared boundary L_{ab}):

$$c_{ab} = \underbrace{\frac{n_a n_b}{n_a + n_b} \|\mu_a - \mu_b\|^2}_{\text{Ward}} \times s_{ab}$$

$$s_{ab} = 1 + 0.5\left(1 - \min\left(1, \frac{L_{ab}}{\min(P_a, P_b)}\right)\right)$$

(s_{ab} favours long shared boundaries; P = perimeter) admissible if same status and $n_a + n_b \le 200$ ha; every round, mutually-best pairs with $c_{ab} < \tau$ merge.

3. Threshold τ — calibrated automatically: 14 geometric bisections in $[10^{-2}, 10^{4}]$ on 300 sample components so that the area-weighted median stand is 15 ha.

4. Minimum mapping unit — then regions < 2 ha merge into their best neighbour (cross-status edges penalised $\times 10^{6}$).

5. Finalise — barrier and sliver pixels join the nearest stand (same status first); fragments < MMU merge by the longest shared boundary.

Outputs — `stands.tif`, polygons with status, area, loss year, TreeMap attribute means, main TreeMap plot (share, count), soil means.

scripts 21_superpixels.py, 22_merge.py, rag_merge.py, 23_finalize.py

HBAD v1 — empirical cohort model (FIA regressions + structural competition)

State per cohort: species, age, TPH, QMD D (cm), height H; site: BA = $\frac{\pi}{4}(D/100)^2$ TPH, SDI = $\sum \text{TPH}\,(D/25.4)^{1.605}$, RD = SDI/$\text{SDI}_{\max}$, BAL = BA of larger cohorts.

$$\Delta D = G_{\text{mult}} \exp\!\left(g_0 + g_1 \ln \tfrac{D}{20} + g_2 \tfrac{D}{20} + g_3 \ln \tfrac{A}{30} - g_k \tfrac{\text{BAL}}{10}\right) \in [0, 3]\ \text{cm/yr}$$

$$H' = 1.3 + (H - 1.3)\,\frac{\text{CR}(D')}{\text{CR}(D)}, \qquad \text{CR}(D) = h_a\left(1 - e^{-h_b D}\right)^{h_c}$$

$$h = M_{\text{mult}} \exp\!\left(m_0 + m_1 \ln \tfrac{D}{20} + m_2 \tfrac{D}{20} + m_3 \text{RD}_c + m_4 \tfrac{\text{BAL}}{10} + m_5 \text{RD}_c \ln \tfrac{D}{20}\right) \le 0.5$$

$$\text{TPH} \leftarrow \text{TPH}\, e^{-h}, \qquad \text{RD}_c = \min(\text{RD}, 1.5)$$

- Reineke self-thinning: while SDI > $\text{SDI}_{\max}$, remove trees from the smallest cohorts up; cohorts end at longevity.
- Establishment: Biomass-Succession block (PROB_ESTAB; light class from RD; new cohort 750 TPH, 1 cm).
- **Fitted**: G_{mult}, M_{mult} (ecoregion × species), $\text{SDI}_{\max}$ (species). **Fixed**: the regression coefficients (FIA GA+FL).
- **Inputs**: FIA cohorts (age, AGB → TPH, QMD; AGB via $\ln m = a + b \ln(D/2.54)$); no climate, CO_2 or soil drivers.

landis_fe docs/hbad_plugin.md, src/plugins/HBAD/plugin.jl

HBAD v2 — biomass-free process cohorts with light competition by height

State per cohort: N (trees/ha), D, H, age A, vigour quantile z. Overtopping by taller cohorts (relative density units):

$$O_i = \frac{\sum_{H_j > H_i} \mathrm{SDI}_j + \frac{1}{2}\sum_{H_j = H_i} \mathrm{SDI}_j + \frac{1}{2}\mathrm{SDI}_i}{\mathrm{SDI}_{\max}}$$

$$\Delta D = r_D\,(D/20)^{c_D}\, e^{-D/D_{\max}}\, e^{-g_{OD} O}\, e^{g_{RD}\mathrm{RD}}\, e^{\sigma_v z}$$

$$\Delta H = r_H\,(H/15)^{c_H}\, e^{-H/H_{\max}}\, e^{-g_{OH} O + g_{RH}\mathrm{RD}}\, e^{\sigma_v z}$$

$$h = f_M\Big[e^{m_{bg} + m_D \ln \frac{D}{20} + m_{D2}\frac{D}{20}} + m_{\sup}\big(1 - \min(1, e^{-g_{OD}O} e^{\sigma_v z})\big)\big(\tfrac{D}{20}\big)^{m_{SD}}\Big] + e^{m_A(A/L - 1)}$$

$$N \leftarrow N\, e^{-h}$$

$$R = 185.2\,\exp\!\big(\rho_0 + \rho_S \ln(1 + \mathrm{BA}_{sp,\,D \geq 12.7}) + \rho_L\,\mathrm{SDI}/600\big)\ \text{trees/ha/yr}$$

- **Parameters**: ≈ 14 per species key ($H_{\max}$, r_H, c_H, $D_{\max}$, r_D, c_D, g_{OD}, g_{RD}, g_{OH}, g_{RH}, m_{bg}, m_D, $m_{\sup}$, $\mathrm{SDI}_{\max}$), globals, recruitment: ≈ 175–210 in all. Recruits enter at 2.54 cm, 3 m.
- **Inputs**: remeasured FIA trees (undisturbed intervals), cohorts by species × 10-yr birth class × size class; **fit**: binomial survival + Gaussian $\ln D$, $\ln H$ of survivors.

landis_hbad2 docs/hbad_v2.md, src/plugins/HBAD2/HBAD2.jl

HBAD3P — process and trait-bound growth (stem-volume balance)

Same cohort state as v2 + crown ratio CR; O, RD as in v2. Water: W = growing-season AET/PET (monthly bucket: PRISM precipitation, Thornthwaite PET, plot AWC); W_n = its plot normal.

$$\Delta V = \varepsilon_k\,(D/20)^b\,20^{1.605}\,\underbrace{\left[f_0 + (1-f_0)e^{-aO}\right]}_{\text{light}}\,e^{g_R \mathrm{RD}}\,\underbrace{W^{\omega}}_{\text{water}}\,e^{\sigma_v z}\left(\frac{\mathrm{CR}}{0.4}\right)^{\varphi}$$

$$\times\,\underbrace{\max\!\left(0,\,1-(H/1.1H_{\max})^{\theta}\right)}_{\text{hydraulic}} - \underbrace{\rho\,D^2H}_{\text{maintenance}}$$

$$\Delta D = \frac{\Delta V}{2DH}$$

$$H^*(D) = 1.37 + (H_{\max} - 1.37)\left(1 - e^{-sD/(H_{\max}-1.37)}\right)^c$$

$$\Delta H = \left[H^*(D+\Delta D) - H^*(D)\right]\left(1+\eta(1-L_h)\right) + \lambda\left(H^*(D)(1+\mu(1-L_h)) - H\right), \quad L_h = e^{-k_h O}$$

$$\mathrm{CR}^* = \mathrm{cr}_{lo} + (\mathrm{cr}_{hi} - \mathrm{cr}_{lo})\,e^{-k_c O}, \qquad \mathrm{CR} \to \mathrm{CR}^* \text{ at rates } \tau_r \text{ (recession)},\ \tau_e \text{ (expansion)},\ \mathrm{CR}_0 = \text{FIA CR}$$

- Per species **traits fixed** ($H_{\max}$, shade tolerance ST, longevity L); per key only the growth efficiency ε_k and fecundity F_k are fitted.

landis_hbad3 docs/hbad3_design.md §4, src/plugins/HBAD3/HBAD3P.jl (step_site!)

HBAD3P — mortality, thinning, recruitment, climate

$$g_m = e^{-a_m O}\, e^{g_R \mathrm{RD}}\, W^{\omega}\, e^{\sigma_v z}\, (\mathrm{CR}/0.4)^{\varphi} \qquad \text{(survival carbon balance)}$$

$$h = f_M\left[\frac{m_0}{L} + m_s\, e^{\sigma_{ST}(\mathrm{ST}-3)}\, e^{-\beta g_m}\left(\frac{D}{20}\right)^{\zeta}\right] + e^{m_A(A/L-1)}$$

$$f_M = e^{p_M[\text{planted}]}\, e^{m_W(W_n - W)/0.1} \qquad \text{(drought anomaly)}$$

$$N \leftarrow N\, e^{-h} \qquad \text{and, while RD > 1: remove SDI} \propto \mathrm{SDI}_i(1 - v_i)$$

$$R_k = F_k\, \exp\!\left(-\lambda_0 e^{\lambda_1(\mathrm{ST}-3)}\,\mathrm{RD}\right) \times [\text{seed source present}]$$

Climate inputs

- growth: W^{ω} (water limitation)
- mortality: drought anomaly $m_W(W_n - W)$
- CO_2, temperature: scenario assumptions from the literature (not identifiable from FIA)

Parameters: ≈ 30–33 per region (vs ≈ 175–210 in v2):

- per key: ε_k, F_k
- growth: b, a, g_R, θ, ω, a_m, ρ, f_0, φ
- height: s, c, η, λ, μ, k_h
- mortality: m_0, m_s, σ_{ST}, β, ζ, m_W
- planted p_D, p_H, p_M; vigour σ_v; crown; recruitment

src/plugins/HBAD3/HBAD3P.jl; docs/hbad3_design.md §4, §7

The three HBAD generations side by side

	HBAD v1	HBAD v2	HBAD3P
growth	FIA ΔD regression $\times G_{\text{mult}}$, BAL	power law in D, H × light (O), crowding (RD), vigour	stem-volume balance: light, crowding, water W^{ω}, crown, hydraulic limit
height	Chapman–Richards ratio	own growth equation	trait H–D curve + light rule
mortality / thinning	regression hazard $\times M_{\text{mult}}$ (RD, BAL); Reineke cap	background + suppression + senescence; SDI cap	carbon-balance hazard + drought anomaly + senescence; RD ≤ 1
recruitment	Biomass-Succession block	stand-table regression	fecundity × light window × seed source
climate	none	none	W (PRISM + AWC), drought anomaly
parameters	G, M per eco×spp, $\text{SDI}_{\max}$	≈ 175–210	≈ 30–33 per region
fit (GA+FL val NLL)	—	−272,655	−273,469

fit: Sim A likelihood (binomial survival + Gaussian ln D, ln H), CMA-ES; HBAD3P free runs: BA / QMD / Lorey H bias within ±3.4% in GA+FL, APP, WSP (docs/hbad3_design.md §8.1)